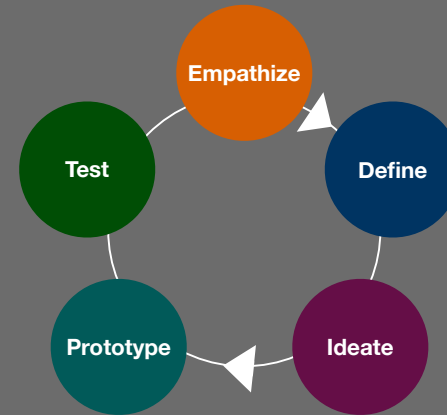
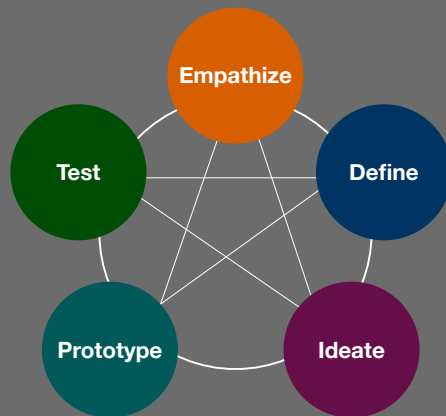


Lecture Three

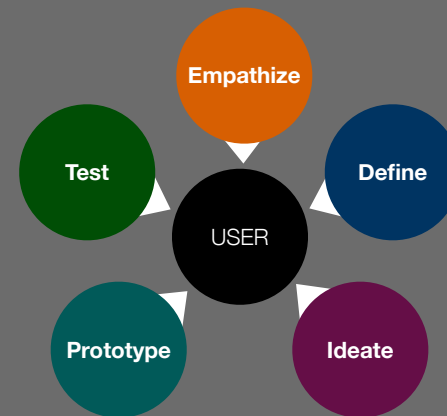
CSCI Technology Innovation | W26
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Design Thinking



Design Thinking



Design Thinking

Product Development Products v Projects

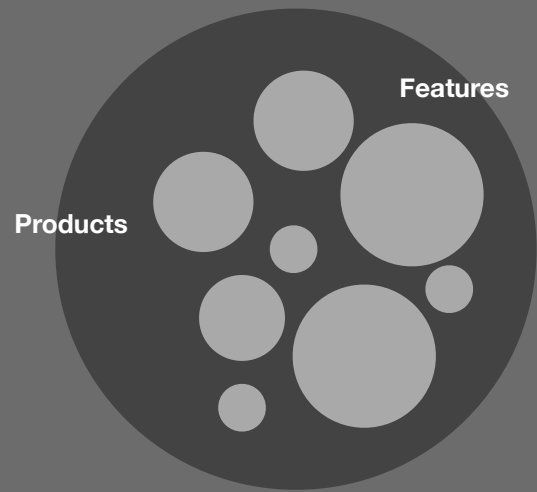
Table 1: Representative differences between projects and products

Project	Product
Solves needs of companies/developers	Solves needs of users/customers
Pre-determined scope	Adapts to needs
Defined beginning & end	No pre-determined end
Success = on time, on budget, intended result achieved	Success = customer satisfaction + ability to evolve

Table 2: Multiple categories of technology products

Hardware	Software	Services	Website	App
Smartphones	Windows/MacOS	Streamers (Netflix, Apple TV+, Hulu)	Search engines	WhatsApp
Laptops	MS Office 365/Google Workspace	Cloud providers (AWS, Azure)	News sites	Zoom
Smartwatches	Web browsers	Online payment systems (PayPal, Stripe)	Online games	TikTok
Gaming consoles	Antivirus software	Webhosting (WordPress, Wix)	Online banking	Instagram
Mouse	Database management systems	Data storage (Dropbox, Google Drive)	E-commerce	Accuweather

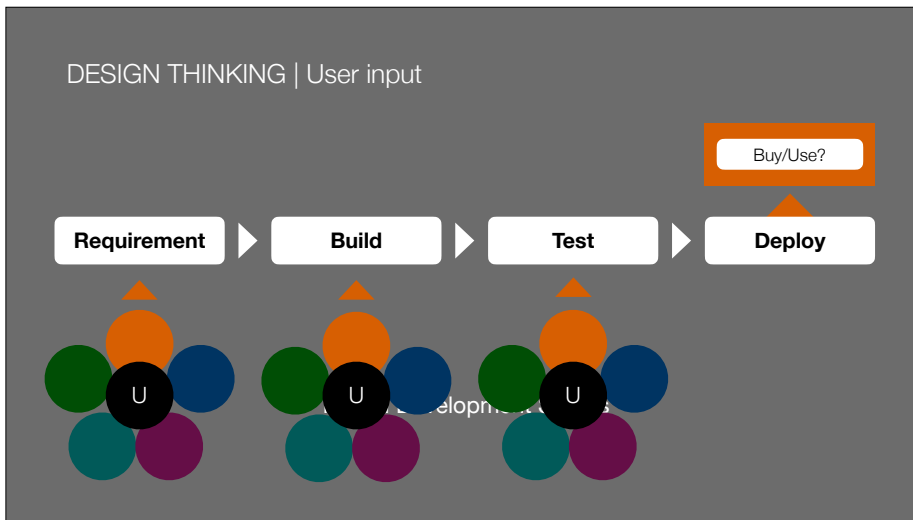
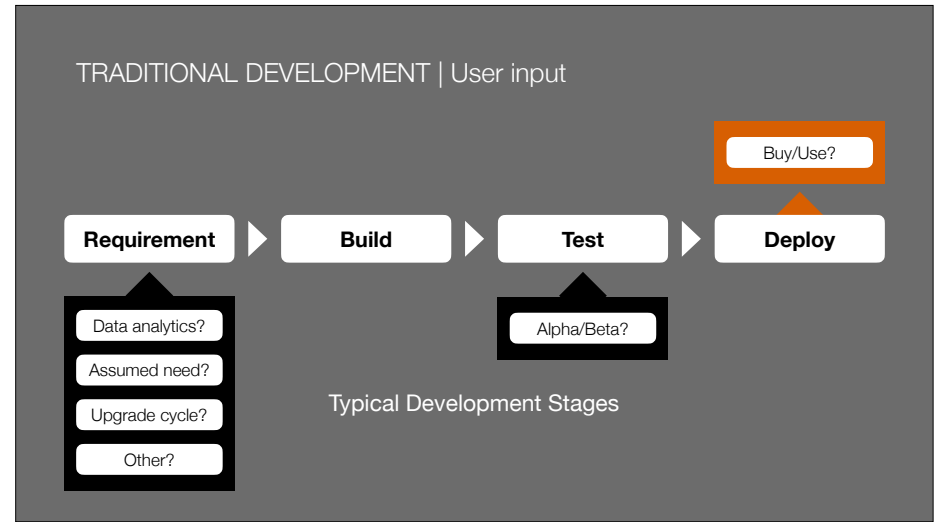
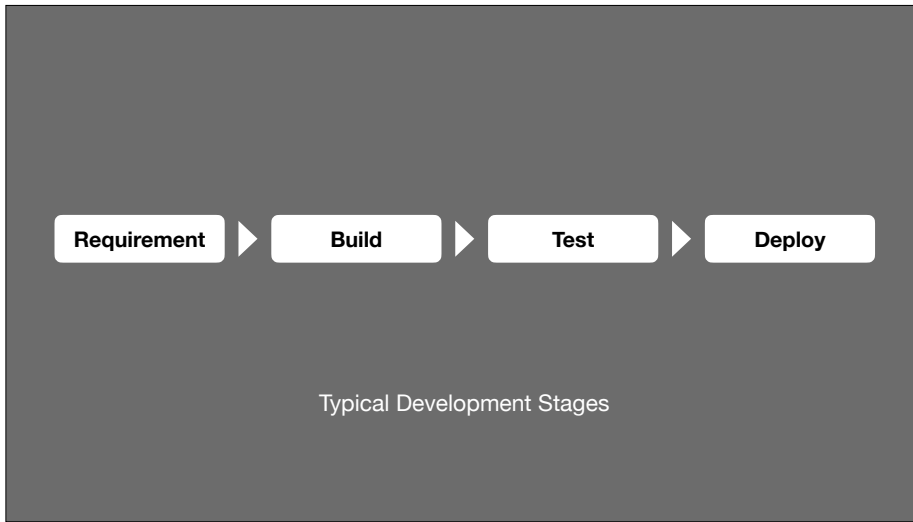
Product Development Products v Features



PRODUCT	FEATURES	PRODUCT	FEATURES
CAR	All wheel drive Multi-speakers ABS brakes AirCon Runflat tires	MS OFFICE	Integration Spellcheck Slide effects Pivot tables Templates
PRODUCT	FEATURES	PRODUCT	FEATURES
RESTAURANT	Menu Ambiance Service Takeaway Desserts	SMARTPHONE	Camera Battery life Screen res Storage Dual sims

We will focus on a project to create a new technology product that offers new and original ways of providing value to a distinct group of users.

Product Development Design Thinking v Traditional Development



	TRADITIONAL	DESIGN THINKING
Time	Long range plan, then predictable process Major issues may not be found until end	Faster start, then step-by-step Issues & opportunities emerge throughout
Resources	Most \$ spent before the results are known Can be high unexpected costs at end	Spend is tied to validated progress Costs rise if many iterations needed
Acceptance	High uncertainty until after launch May be significant post-launch changes	Higher confidence when launched Faster adoption & fewer adjustments



Understanding a user's desires while figuring out the product

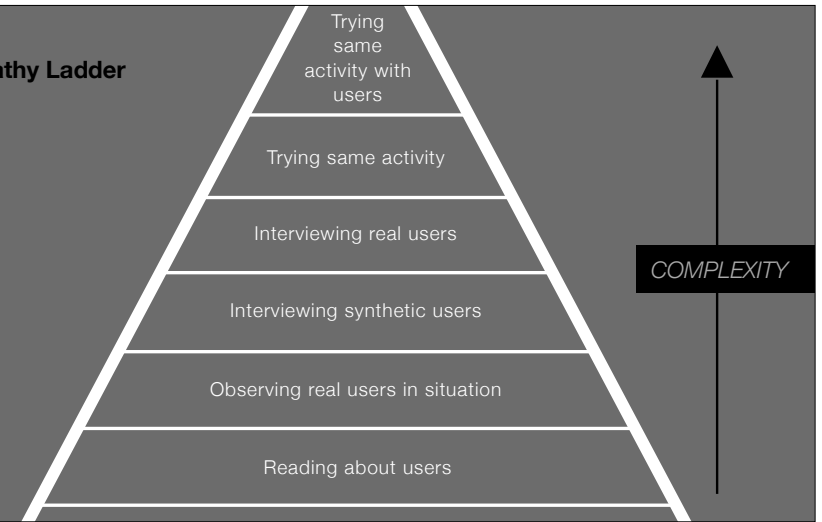
"Having the questions before the exam"

Suspending judgment, having respect for user's experience & desires

Pure curiosity

Customer Discovery

Empathy Ladder



Asking good questions

Open v closed (*Why? How? When?*)

Mix of structured v unstructured

Creating a conversation

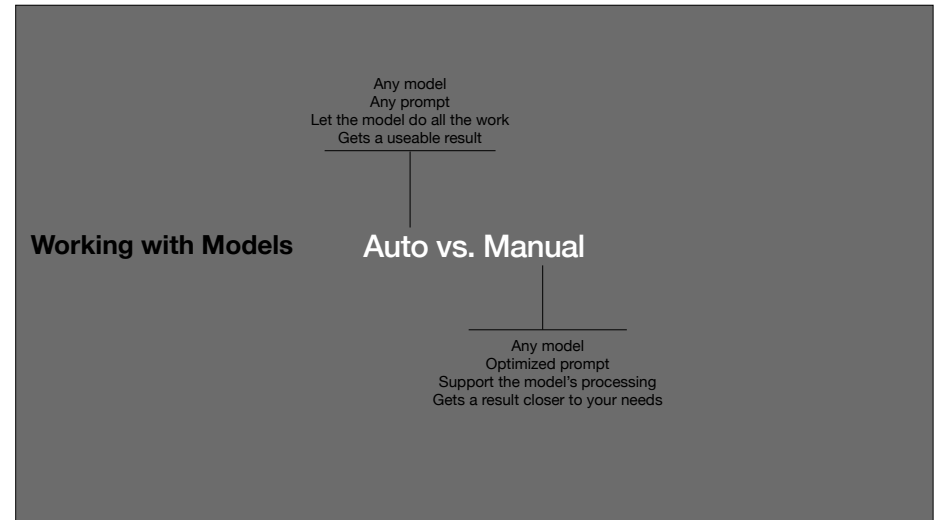
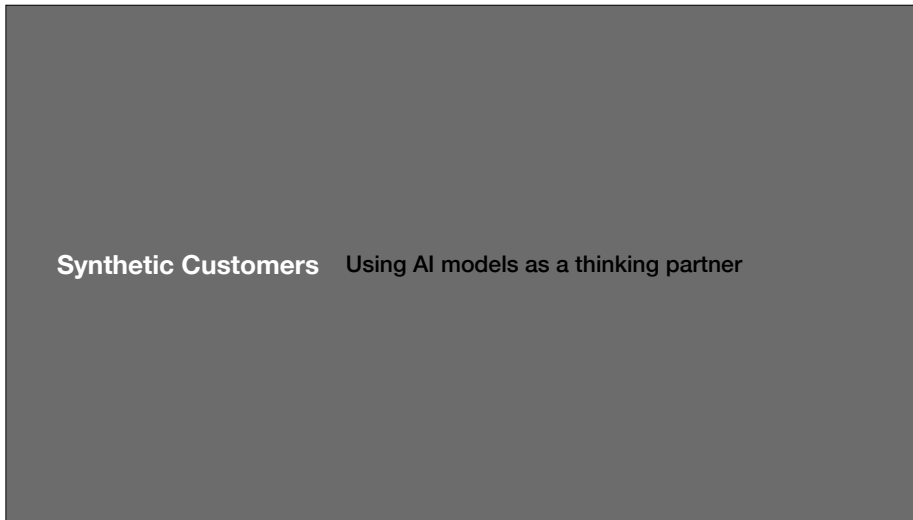
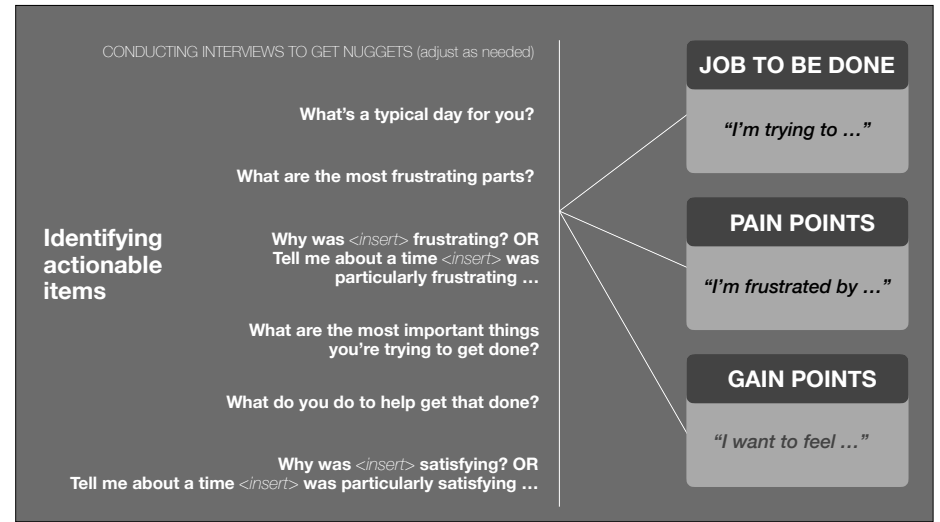
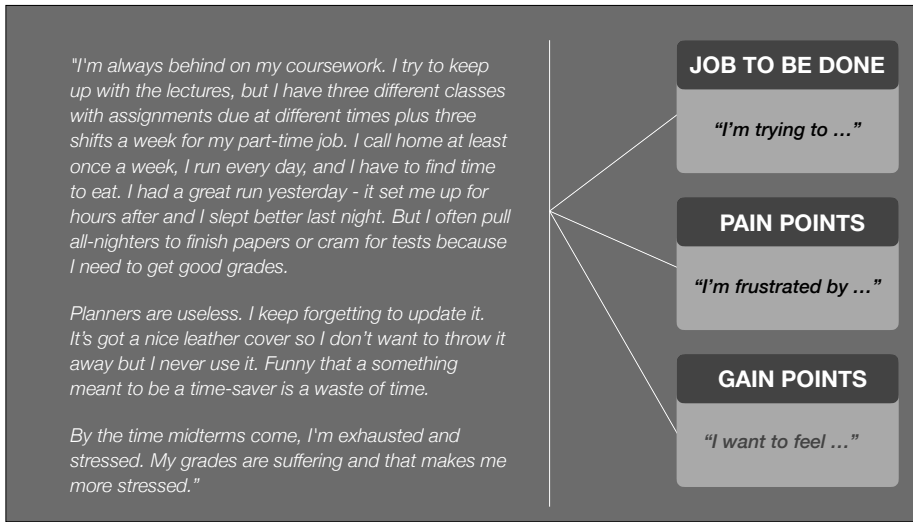
Encouraging stories (*Tell me about a time when...*)

Identifying actionable items

Nuggets v noise

Reveals desire
Specific example

Vague/general
Off-topic



Using three "dials" to fine-tune your prompt in "manual mode"

Context

Constraints

Clarity

How much background will be helpful?

Context

The model processes only from the information it has

- training data
- supplementary data, e.g. from RAG or WWW
- your prompt

The prompt provides essential "just-in-time" extra training for the response you want to get

For humans, context = background
For models, context = training

How can limitations improve the result?

Constraints

The model has a vast capacity to generate information

The prompt benefits by focusing on the specifics you want. For example:

- length/format?
- level of comprehension for audience?
- tone?
- things to include?
- things to avoid?
- etc.

Are the instructions as clear as they can be?

Clarity

Ambiguity is the enemy of generation

Models are left to "guess" the meaning for words like

- explain
- analyze
- good
- simple
- shorter... longer ... more ... less ...

Checking the best word choices in this instance

Three factors for optimizing prompts

CONTEXT What information will support the model's work on my query?

CONSTRAINTS What boundaries will focus the results I receive?

CLARITY How can I make my meaning more obvious to the model?

THE BIG IDEA

To get more tailored results from a language model, switch to “manual mode” when prompting the model - use *context*, *constraints*, and *clarity* to fine-tune the model's processing of your request.

**Things to
check & avoid**

Tarpit ideas

High effort but limited progress (and likely abandoned by others)

“Paving the cow path”

Digitizing an existing process without questioning if it's useful

Confirmation bias

Favouring ideas/evidence because they align with your POV

Ideas without 3rd-party evidence

Trusting your product ideas are strong without external validation

THE BIG IDEA

When developing new products, replace assuming with learning. Frequent checks with users, throughout the process, can save money and increase customer satisfaction.